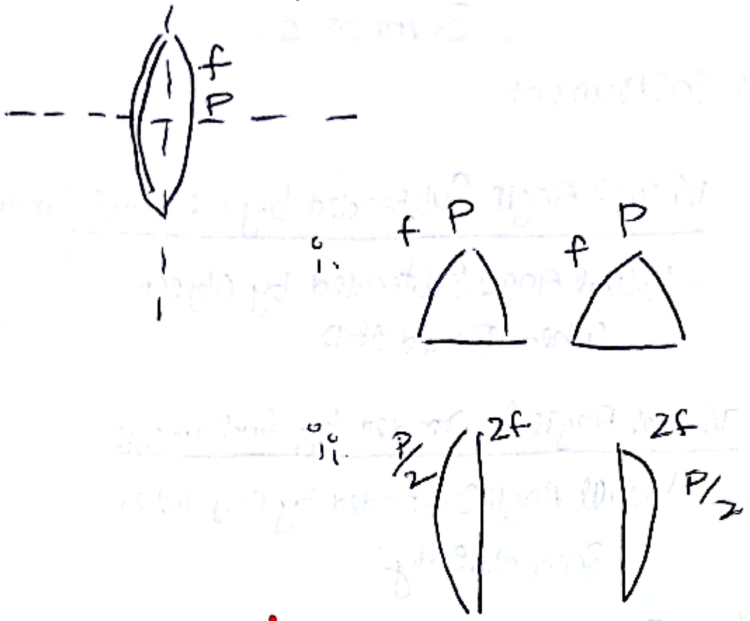
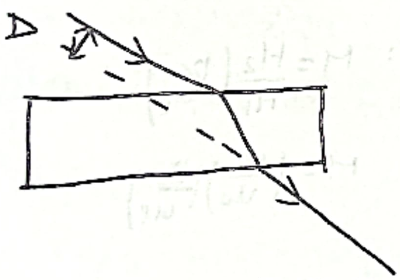
⇨ CUTTING OF LENS:



RAY OPTICS...



$\Delta \rightarrow$ Lateral Displacement.

$$\Delta = \frac{t\sin(i-r)}{\cos r}$$

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*

Synopsis

Optical instrument:

$$M = \frac{\text{Visual Angle Subtended by the final image}}{\text{Visual Angle Subtended by Object When Placed at D}}$$

$$M = \frac{\text{Visual Angle Subtended by final image}}{\text{Visual Angle Subtended by Obj when Seen directly}}$$

Magnifing Power $M = \frac{\theta}{\theta_0}$

$M = \frac{D}{u}$

Case I:

$M = \frac{D}{f}$

Case II:

$M = \left(1 + \frac{D}{f}\right)$

$u = \frac{Df}{D+f}$

For Compond Microscope: $M = \frac{H_2}{H_1}\left(\frac{D}{u_e}\right)$

$M = \left(\frac{v_o}{u_o}\right)\left(\frac{D}{u_e}\right)$

For length Microscope:

$L = v_o + u_e$

$M = \left(\frac{v_o}{u_o}\right)\left(\frac{D}{u_e}\right)$

Case 1:

$L = v_o + f_e$ & $M = \left(\frac{v_o}{u_o}\right)\left(\frac{D}{f_e}\right)$

Case 2:

$L = v_o + \frac{Df_e}{D+f_e}$

SUB TOPICS

- Power of Lens & Mirror.
- Reflection of light.
- Laws of reflection. $M = \frac{f}{f-u}$
- Finding Total deviation.
- field of View $r_{Min} = \frac{h}{\sqrt{\mu^2-1}}$
- Min Height of Mirror
- cases in concave convex. $(P)_{Mirror} = \frac{(-1)}{f}$
- Snells Law.
- Refractive index.
- Total internal refraction.
- Appications of TIR. (Mirage, Looming, Optical fiber)
- Finding Normal Shift due to Slab.
- Prism Theory
- Minimum Deviation.
- lateral displacement formula $(d) = \frac{t\sin(i-r)}{\cos r}$.
- Thin Prism.
- Condition for no emergent ray: $\boxed{r_1 = A/2}$
- Lens Maker formula $\sin^{-1}\left(\frac{3}{4}\right) = 49^\circ$

$$\frac{1}{f} = \left(\frac{\mu_2}{\mu_1}-1\right)\left[\frac{1}{R_1}-\frac{1}{R_2}\right]$$

- $\frac{1}{v}-\frac{1}{u} = \left(\frac{\mu_2}{\mu_1}-1\right)\left[\frac{1}{R_1}-\frac{1}{R_2}\right]$
- $f_{water} = 4f_{air}$.
- Displacement Method to find focal length.
- Transverse Magnification.
- Lens & Lens Maker Formula $\boxed{\frac{\mu_2}{v}-\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R}}$ ✱
- Finding focal length of lens

$$\frac{1}{f} = \left(\frac{\mu_2}{\mu_1}-1\right)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$ ✱

- Dispersion of light $\boxed{\mu = \frac{1}{\lambda}}$ $\mu = A + \frac{B}{\lambda^2}$

FORMULAS

KEY CONCEPTS

: $\hat{R} = \hat{I} - 2(\hat{I}\cdot\hat{N})\hat{N}$

: deviation $\delta = 180 - 2i$

: Min height of Mirror $= \frac{h_{person}}{2}$

: Only $\hat{i}$ sign will change Both $\hat{j}\hat{k}$ not change in case of Velocitys of Obj and Images

: $V_{IO} \Rightarrow$ Vel of image w.r.o Obj.

$\boxed{V_{IO} = V_I - V_O}$

: If Mirror is rotated through an angle θ then I should rotate 2θ Same course with Velocity's

: No. of images formed

$$= \left(\frac{360^\circ}{\theta^\circ} - 1\right)$$

: Ray after reflection from 3 ⊥er Mirror's becomes Anti Parallel

: $\frac{dv}{du} = -\left(\frac{v}{u}\right)^2 \Rightarrow \boxed{\frac{dv}{dt} = -\left(\frac{v}{u}\right)^2 \frac{du}{dt}}$

*** Formula for Mirror: $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

: Magnification (M) $= \frac{h_i}{h_o}$

: focal length doesn't change whether the Mirror is in air (or) immersed in water !

: freq doesn't change under μ

: Power of Plane Mirror is Zero

: App rise $= d\left(1 - \frac{1}{\mu}\right)$

: $\delta = (\mu - 1)A$

** Jee Pyq.

Snell's law: $\mu_1 \sin\theta_1 = \mu_2 \sin\theta_2$ (μ_1) < (μ_2)
rarer denser

$\boxed{\mu \sin\theta = \text{constant}}$

** : When ray enters b/w n No. of μ values the final emergent ray is Also Parallel to incident

→ Apply's for 1st last Surfaces

Apparent depth = $\frac{\text{Real depth}}{\mu}$.

$$\boxed{\Delta = \frac{t \sin(i-r)}{\cos r}} \quad **$$

⇨ $A_I = \left(\frac{\mu_1}{\mu_2}\right) A_0 \; (A_I < A_0)$

⇨ Vel of image in transverse direction

(Ler to Pricipal Axis $= -\left(\frac{v}{u}\right)$ Transverse Speed direction

⇨ Vel. of image in longitudinal dir

(Along Pricipal Axis) $= -\left(\frac{v}{u}\right)^2$ Longitudinal Speed dir.

*** Apparent depth = $\frac{\text{refractive index of denser}}{\text{refractive index of rarer}}$ (real depth).

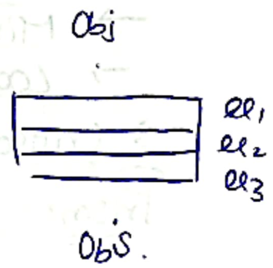
** if n no. of Parallel Slabs are b/w Obs and Obj then total app shift to

$$S = \left(1-\frac{1}{\mu_1}\right)x_1 + \left(1-\frac{1}{\mu_2}\right)x_2 + \left(1-\frac{1}{\mu_3}\right)x_3 + \ldots + \left(1-\frac{1}{\mu_n}\right)x_n.$$

Where

$x_1 x_2 x_3 - - x_n$ are thickness of slab.

$\mu_1 \mu_2 \mu_3 - - \mu_n$ are refractive index.



: if Point Obj is kept at the first focus of a convex lens if lens starts moving towards right with constant vel. image first moves to left and then move towards right.

: Apparent dist = $\mu_{\text{w.r.t. other}} \times \left[\frac{\text{dist}_1}{\mu_1} + \frac{\text{dist}_2}{\mu_2} + \ldots + \frac{\text{dist}_n}{\mu_n}\right]$ ✓

: Normal Shift :

$$X = \left(1-\frac{1}{\mu}\right) \text{Thickness of Sub}.$$

where δ angle is zero !

//

* $\mu = \dfrac{\text{Speed of light in Vaccum (Air)}}{\text{Speed of light in Water (Medium)}} = \dfrac{c}{v}$

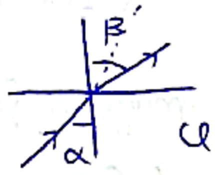
⇨ μ (diamond) = 2.4.

⇨ θ_c (critical angle) = $\sin^{-1}\left(\dfrac{\mu_1}{\mu_2}\right)$

$$\boxed{{}_i\mu_j = \frac{\mu_j}{\mu_i}}$$

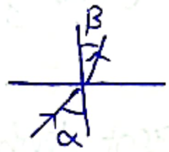
⇨ From Denser to rarer :

$\delta = \beta - \alpha$.

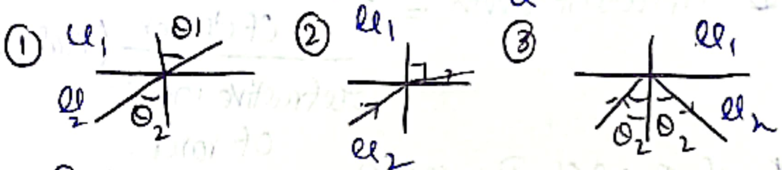


⇨ From rarer to Denser :

* $\delta = \alpha - \beta$.



* Cases in TIF :



① $\theta_2 < \theta_1$, $\mu_2 > \mu_1$

② $\theta_c = \theta_2$

③ $\theta_2 > \theta_c$. $\boxed{\text{TIR}}$

→ Optical fibres
→ Diamond Sprakle
→ Mirage
→ Looming

* Towards N : rarer - Denser
Away N : Denser - rarer

: Critical angle increased relative μ decreases

Prism:

- $r_1 + r_2 = A$
- $\delta = (i_1 + i_2) - A$.
- * (i) $\sin i_1 = \mu \sin r_1$
  (ii) $\sin i_2 = \mu \sin r_2$ } By Snells Law
- TIR, Denser to rarer.

Formulas

For Min deviation

* $d = \dfrac{t\sin(i-r)}{\cos r}$

* $d = \dfrac{t(i-r)}{1}$ (small angle)

$\mu = \dfrac{\sin\left(\frac{A+\delta_{Min}}{2}\right)}{\sin(A/2)}$ * **

$i_1 = i_2$

$r_1 = r_2 = A/2$ $\boxed{\mu = \dfrac{\sin i_1}{\sin r_1}}$

* $\boxed{\delta_{Min} = 2i_1 - A}$

$\delta = A(\mu - 1)$ when $A \simeq 0°$

Dispersion of Light :

$$\boxed{\mu = A + \frac{B}{\lambda^2}} \quad (1)$$

⇒ Cauchy relation.

$\boxed{\text{VIBGYOR}}$

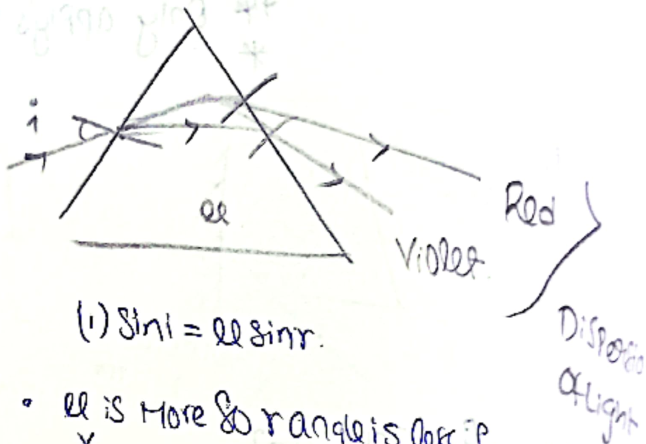
⇒ (V) wavelength < (R) wavelength

⇒ Hence using eq (1) $\boxed{\mu_V > \mu_R}$

⇒ Refractive index of all colours is not same.

**

	μ.
V	1.52
I	1.48
B	1.46
G	1.44
Y	1.42
O	1.41
R	1.4



(1) $\sin i = \mu \sin r$.

- μ_V is more & r angle is less for Violet

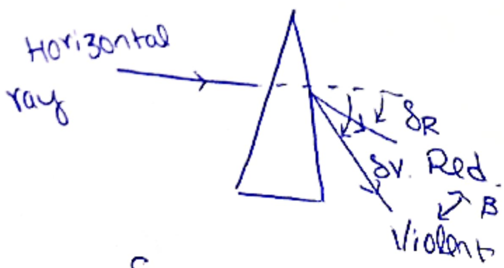
- In Parallel Slab there is no change (May) of Dispersion of light in Prism we know, change of Dispersion exist !!!

Hence

** By Prism Red, Violet is been separated ..

**

Small Angled Prism.



$\delta = (\mu - 1)A$.

$\delta_V = (\mu_V - 1)A$ & $\delta_R = (\mu_R - 1)A$.

	M	L	Case ①	Case ②
Simple	$\frac{D}{u}$	—	$u=f$	$u=\frac{Df}{D+f}$
compound	$\left(\frac{v_o}{u_o}\right)\left(\frac{D}{u_e}\right)$	v_o+u_e	$u_e=f_e$	$u_e=\frac{Df_e}{D+f_e}$
Astronomical	$\left(\frac{f_o}{u_e}\right)$	f_o+u_e	$u_e=f_e$	$u_e=\frac{Df_e}{D+f_e}$
Terrestrial	$\left(\frac{f_o}{u_e}\right)$	f_o+4f+u_e	$u_e=f_e$	$u_e=\frac{Df_e}{D+f_e}$
Galileoun	$\left(\frac{f_o}{u_e}\right)$	f_o-u_e.	$u_e=f_e$	$u_e=\frac{Df_e}{D-f_e}$

Where.

usually 25cm

D → Least distance of distinct Vision

u → Object distance

u_e → Object distance from eye piece

f → focal length

f_e → focal length of eye piece

u_o → Object distance for objective lens.

V_o → Image distance from Objective lens.